

TEST & MEASUREMENT

An Innovation A Day!

The T&M world is innovating at a rapid pace, to keep in step with the developments taking place in its client industries, ranging from communications and computing to aerospace and defence. Radar technologies, 3-D imaging, increasing precision, unbelievable speeds, network convergence, wireless capability, video tests, virtual instruments—catch a glimpse of all these and more trends in T&M

■ JANANI GOPALAKRISHNAN VIKRAM

If there is one factor more important than innovation, it is probably quality. The competitive environment in every industry has ensured that customers never have to compromise; they can choose the best and ignore the rest. To be the best, a product has to be designed, developed and manufactured not just fast but also perfectly—a task that is impossible without the latest test and measurement (T&M) equipment.

Consequently, engineers and manufacturers want the best equipment, be it for compliance testing, communication testing solutions, machine vision and inspection, and so on. "Compliance testing, communication testing and aerospace/aviation/defence equipment have the highest demand in India, though almost all the sectors have their own requirements and demands. This is because these segments have been developing at a rapid pace and significant technological investments are happening in the Indian aerospace, aviation and defence sector," comments Sai Venkat Kumar of Tektronix's Asia-Pacific Marketing Group.

Today, as we know, the business equation is more than just 'demand→supply.' It is now 'demand→innovation.' This has led the T&M industry to come out with instruments that are not only more precise but also imbibe all the latest trends ranging from wire-

less networking to three-dimensional (3D) imaging. Here is a small sample of the latest technology available in the T&M space.

Green T&M helping to go green

Considering that green is the colour of the day, let us begin by looking at how the current generation of test and measurement equipment aids engineers in designing and developing eco-friendly,

energy-efficient electronic products.

To be able to design products that consume less energy, engineers must be able to measure the energy consumed at not only the system level but also at the component level, because every milliwatt matters. This is a high-precision T&M task. For example, when working with components such as light-emitting diodes and power semiconductors, engineers need low-level measurements such as forward and reverse voltage

Innovative Products Launched Last Year

EFY Bureau team member spoke to the leading T&M companies in India to get an idea about their most innovative launch last year. Here are the chosen ones...

Fluke 233 DMM with Detachable Display

One of the biggest challenges in some of the measurements is that the point of measurement is at a location where you cannot conveniently view the readings—say, underneath an automobile. Detachable display is a perfect solution for such measurements.

Fluke's 233 wireless remote display multimeter is world's first such device that allows you to be 10 metres away from the measurement point. By simply sliding the wireless display

out of the meter body, the user can place the display where it can be seen most easily, while putting the meter in the best position to take measurements.

This multimeter uses a 2.4GHz ISM band wireless transmitter to immediately send measurements to the detached display. Designed for both convenience and safety, it makes it easier to take measurements in hard-to-reach places, or make measurements in machines and panels that are physically separated from a limit or isolator switch. (Distributed in India by TTL India (www.ttlindia.com))

